GCI World 2026 September, final assignment

Keeping customers before they leave

A churn-risk retention proposal for Company A, a telecom operator

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Contact the customers most likely to leave, and do it before they do

70%

of the highest-risk 20% of customers churned, against 38% in the lower half

58%

churn at 10–12 months of tenure, against 31% at 7–9 months: a renewal window

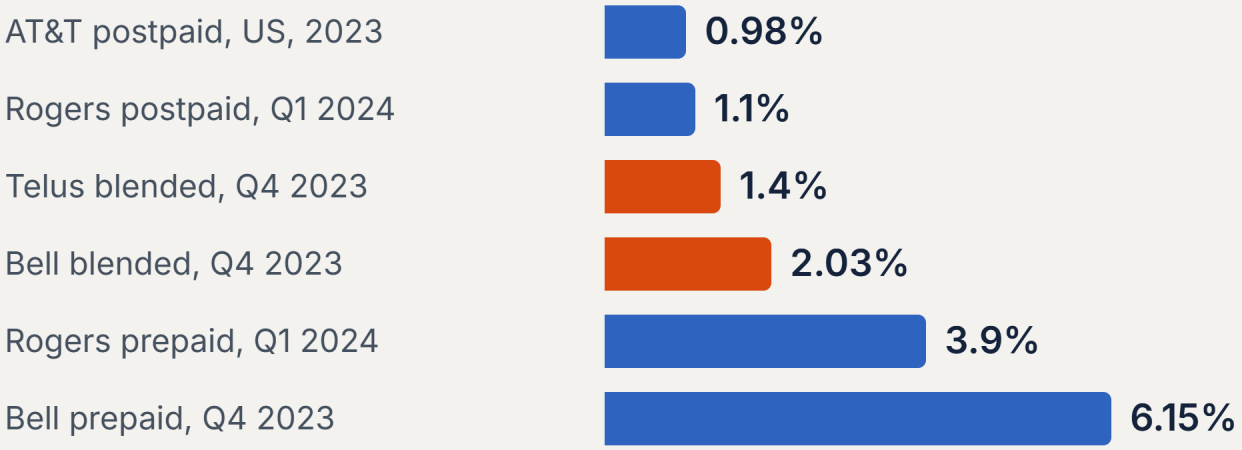
~14,600

net benefit per 100,000 customers in one month of churn, under stated assumptions

Proposal: rank customers with a random forest (ROC-AUC 0.661), contact the top 10% through low-cost channels at the right moment, and prove the effect with a holdout pilot.

Mobile operators lose about 1–2% of subscribers a month, and prepaid churn runs several times higher

Monthly churn at large mobile operators (orange = blended postpaid and prepaid)



Keeping a customer is usually at least **5×** cheaper than winning a new one [10].

Global mobile churn averages under 3% a month. At 3% an operator replaces its whole base in about three years [11].

Company A shared 100,000 customers and 100 columns, and about half of them churned

100,000

customers in two tables, joined one-to-one on Customer_ID

100

columns: account, handset, usage, call quality and tenure

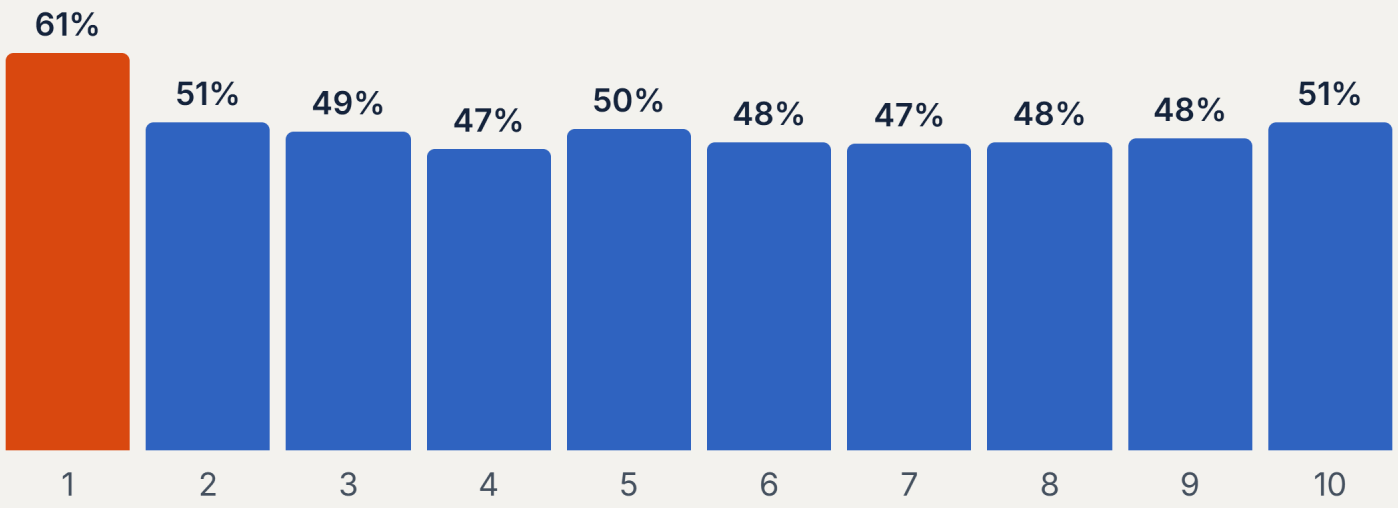
49.6%

of customers left 31–60 days after the observation date

- No duplicate customers or rows after the merge, and a clean yes/no churn label.
- Household fields (income, dwelling and similar) are empty for 22–49% of customers, so we left them out.
- 2,839 newer customers lack six-month history: we flag them instead of filling zeros.
- Real churn is 1–2% a month, so this sample is probably balanced on purpose: its rates describe the sample, not the whole base.

Falling usage predicts churn only when the drop is steep: the steepest 10% churn at 61%

Churn rate by decile of recent change in voice usage (1 = steepest drop)



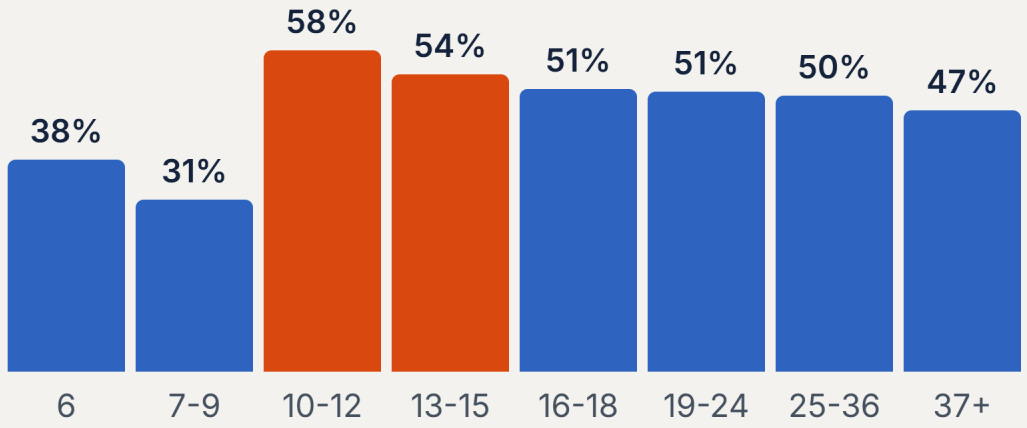
61% churn among customers whose recent voice minutes are at least a third below their six-month level (overall: 49.6%).

75% churn among the 1,761 customers (1.8%) with no voice usage at all in the last three months.

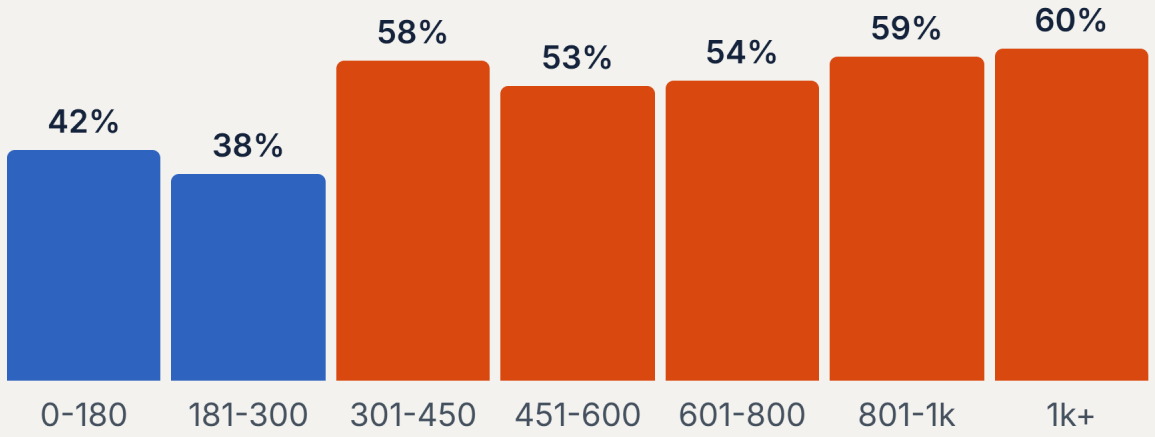
Mild declines carry no signal: deciles 2–10 sit at 47–51%.

Churn jumps at about one year of tenure and once the handset is more than 300 days old

Churn rate by tenure (months)



Churn rate by handset age (days; k = 1,000)



Churn is 31% at 7–9 months and 58% at 10–12 months, then eases to 47% beyond three years. It looks like a contract-end or upgrade window, but the data does not mention contracts, so we treat it as a hypothesis to test.

We rank customers by churn risk, so the retention team knows who to contact first

Customer data
Two tables, 100,000 customers

Features
Tenure, handset age, usage levels, usage decline

Model
Estimates each customer's chance of churning

Risk ranking
Top 20%, next 30%, lower 50%

Retention action
Low-cost offers, tested against a holdout

Business goal
Put retention spend where it is most likely to change an outcome, instead of discounting everyone.

Machine learning task
Classify whether a customer churns (leaves 31–60 days after the observation date). 80/20 train–test split, stratified. Main metric: ROC-AUC, because the use is ranking.

A random forest ranks churners best: ROC-AUC 0.661 on 20,000 held-out customers

Model	ROC-AUC	Accuracy	Precision	Recall	F1
Random forest (selected)	0.6609	0.6112	0.6031	0.6302	0.6164
Gradient boosting	0.6545	0.6050	0.5925	0.6503	0.6201
Logistic regression	0.5902	0.5661	0.5635	0.5527	0.5580

What the score means

Pick one churner and one non-churner at random: the model ranks the churner higher 66% of the time (0.5 is a coin flip).

Why a random forest

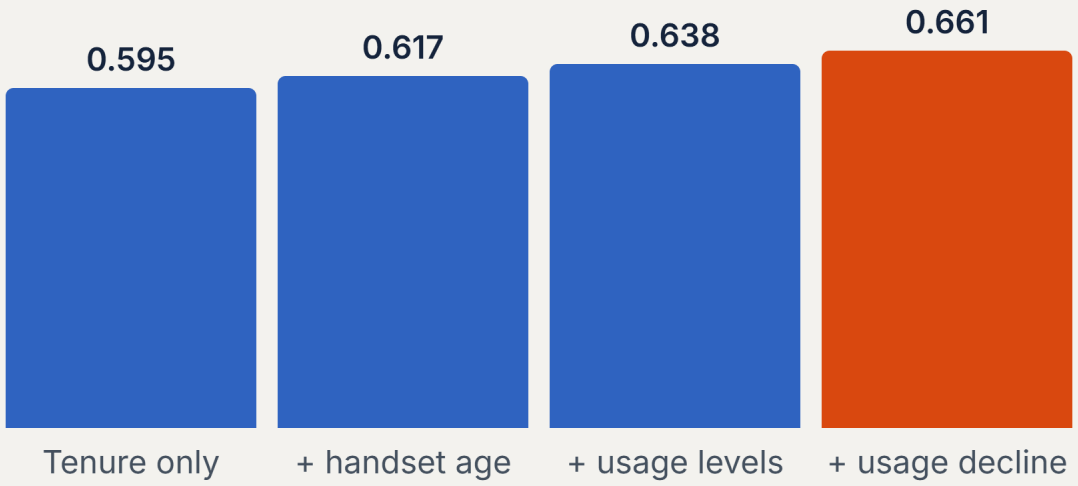
Usage decline works as a threshold, which logistic regression (0.590) cannot capture. Trees can.

Honest read

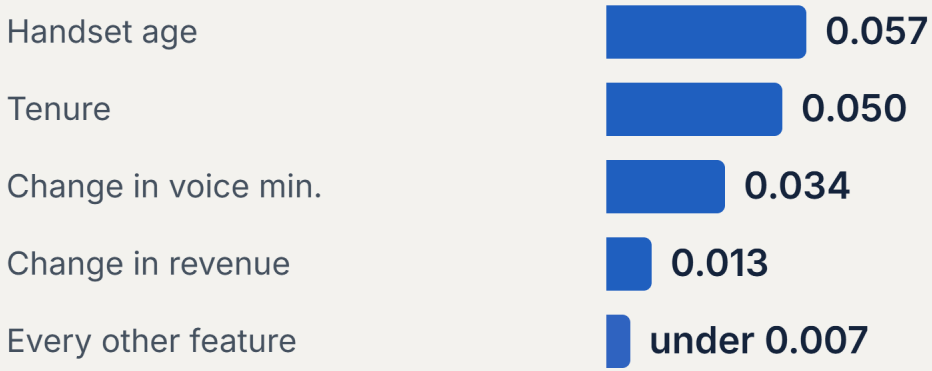
Churn is only partly predictable from these variables. Use scores to prioritise, not to decide one customer’s fate.

Tenure and handset age lead, and usage decline adds a steady extra lift

ROC-AUC as feature groups are added



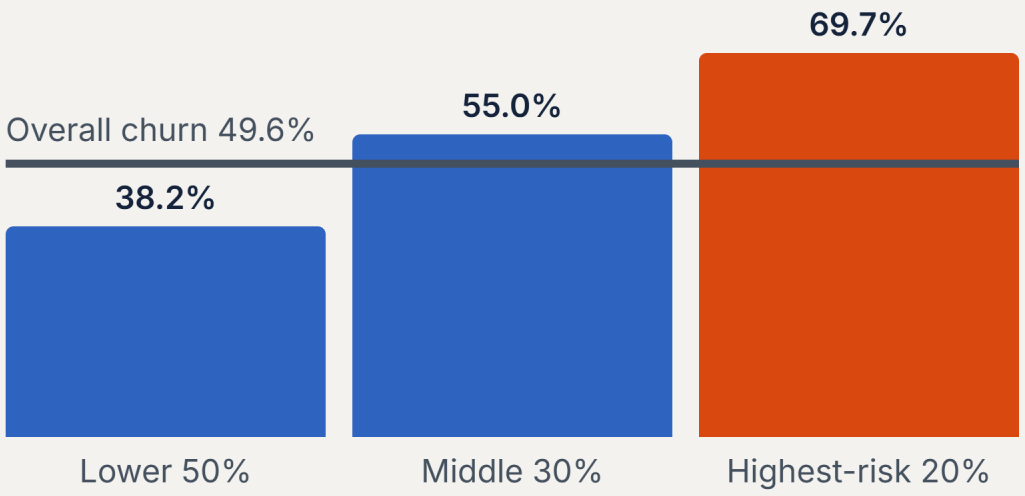
Feature importance (drop in ROC-AUC when the feature is shuffled)



Five-fold cross-validation agrees: ROC-AUC rises from 0.637 (± 0.004) to 0.658 (± 0.003), and decline helps in every fold. These are associations, not causes.

The top 20% by predicted risk churn at about 70%, against 38% in the lower half

Observed churn by predicted-risk group (test set)



Group	Median tenure	Change in voice minutes	Failed calls
Lower 50%	16 months	+4	5.5%
Middle 30%	19 months	−9	5.4%
Highest-risk 20%	13 months	−43	5.6%

The highest-risk group sits near the one-year mark with falling voice usage. Failed-call rates are the same in all three groups, so network quality does not explain who is at risk here.

Risk-ranked retention, timed around the renewal window

Contact first

Customers whose voice usage fell by a third or more, or who had no usage in three months.

Renewal window

Customers at 10–15 months of tenure, when churn is highest.

Handset offers

Test upgrade offers on handsets older than about 300 days (a hypothesis to test).

Delivery: low-cost automated channels to the top 5–10% by risk. **Proof:** a pilot with a holdout group. **Not recommended:** a service-quality programme, since failed-call rates do not separate churners.

The impact estimate rests on conservative assumptions drawn from market data

Real monthly churn	**2%**	Blended churn is 1.4–2.0% at Telus and Bell, postpaid is 1.0–1.6%, prepaid 3.9–6.2% [1]–[4].
Contribution margin	**30%**	Median telecom EBITDA margin is about 34%, so 30% is deliberately lower [5][6].
Save rate	**10–30%**	Targeted campaigns cut churn from 68% to 50% in a high-risk group [7]; a B2B case saw 6% against 34% in controls [8]. Case studies, so we test it.
Cost per contact	**0.50**	A text message costs roughly $0.01–$0.05 [9]; 0.50 leaves room for offer credit or the odd agent call.
Saved customer stays	**12 months**	Our assumption; to be replaced with Company A's own lifetime data.

At 0.50 per contact, the campaign nets about 14,600 per 100,000 customers in a month

10,000
customers contacted (top 10%)

~116
churners saved at a 20% save rate

~19,600
margin retained

5,000
campaign cost at 0.50 each

~14,600
net benefit

Break-even cost per contact (the most one offer can cost before the campaign loses money)

Target share	Save rate 10%	Save rate 20%	Save rate 30%
Top 5%	1.12	2.24	3.36
Top 10%	0.98	1.96	2.94
Top 20%	0.87	1.75	2.62

At 2% real churn, precision among contacted customers falls from about 80% in the sample to 5–8%. So the model supports cheap, automated outreach to the top 5–10%, not big discounts.

A holdout pilot will measure the real effect before any wider rollout

What could go wrong

- The sample churns at about 50%, so the impact relies on assumed real churn, margin and save rate.
- Only 3- and 6-month averages exist, and the two windows overlap.
- ROC-AUC 0.661 is modest, and results are associations, not causes.
- The top two models are close and were compared on one test split.

Proposed pilot

1. Score all customers every month.
2. Contact the top 10% by risk, and hold back a random control group.
3. Compare churn and margin against the control group over 8–12 weeks.
4. Scale up only if the net benefit is positive at the real cost per contact.

Sources and tools

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